

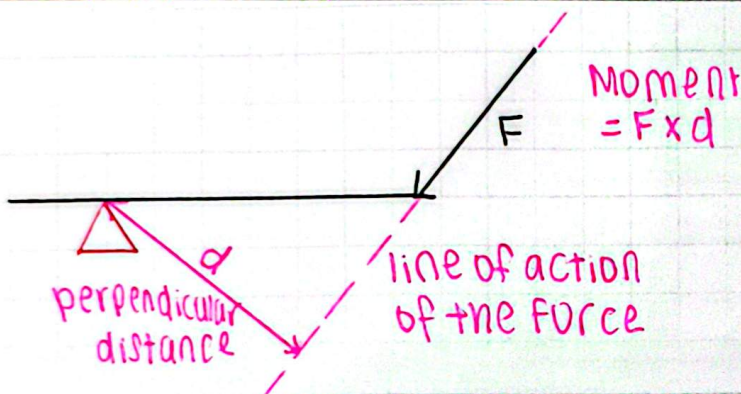
Chapter 5. Turning Effects of Forces

Moment of a Force (definition)

- The moment of a force about a pivot is the product of the force and the perpendicular distance from the pivot to the line of action of the force.

$$\text{Moment (Nm)} = \text{Force (N)} \times \text{Perpendicular Distance (m)}$$

$$\text{Moment} = F \times d$$



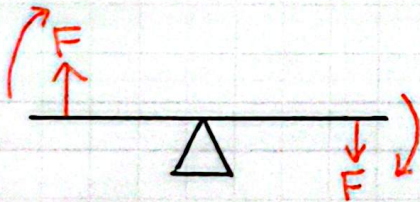
$$\text{Moment} = F \times d$$

Moment is a vector quantity.

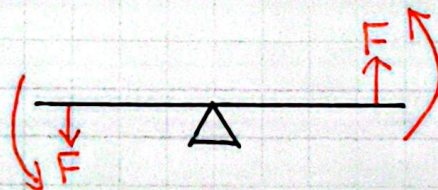
- consist of magnitude and direction (clockwise / anti-clockwise)

Direction of moments

Clockwise moment



Anti-clockwise moment



Zero Moment

- When the line of action of the force passes through the pivot, there is no moment (zero distance)

Principle of moments (definition)

- When a body is in equilibrium, the sum of clockwise moments about a pivot is equal to the sum of anti-clockwise moments about the same pivot.

$$\text{Sum of clockwise moments} = \text{Sum of anti-clockwise moments}$$

(about pivot P) (about pivot P)

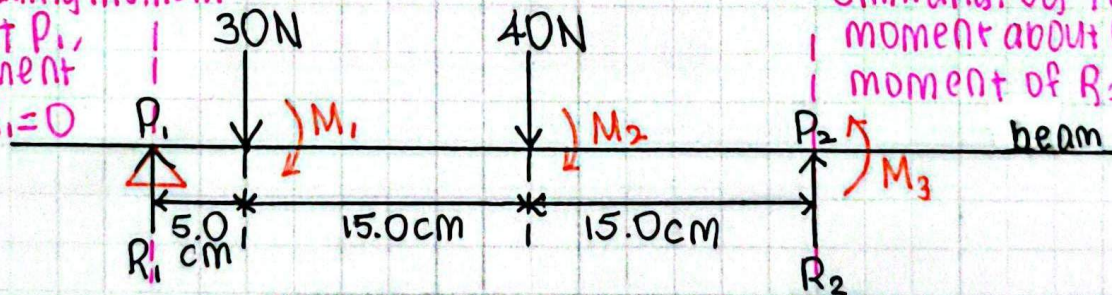
Forces in equilibrium (definition)

- For a body to be in equilibrium, the resultant force must be equal to zero.
- Therefore, if body in equilibrium,
total upward forces = total downward forces

Finding unknown forces using principle of moments

Example: Find the magnitude of R_1 and R_2 if the beam is in equilibrium.

By taking moment about P_1 ,
moment of $R_1 = 0$



Similarly, by taking moment about P_2 ,
moment of $R_2 = 0$

Using Principle of Moments, taking moment about P_1 ,
sum of clockwise moments = sum of anti-clockwise moments
 $M_1 + M_2 = M_3$

$$30 \times 5.0 + 40 \times 20.0 = R_2 \times 35.0$$

$$R_2 = 27 \text{ N (2 s.f.)}$$

Since the body is in equilibrium,

Total Upward Forces = Total Downward Forces

$$R_1 + R_2 = 30 + 40$$

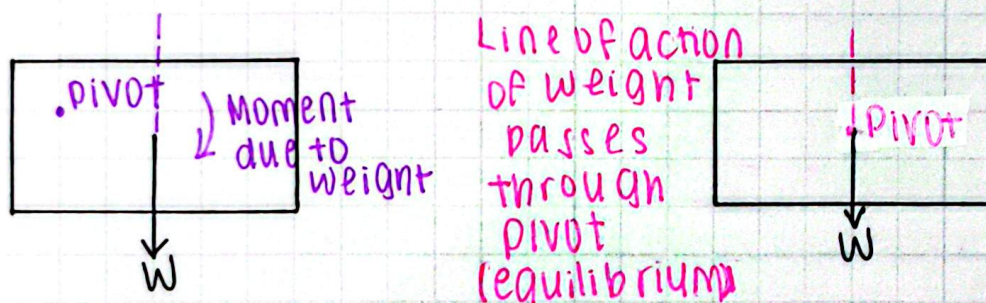
$$R_1 = 43 \text{ N}$$

Centre of gravity (definition)

- The centre of gravity is an imaginary point where the entire weight of the object seems to act.

If the line of action of the weight of an object does not go through the pivot, a moment exists that makes the object turn.

The object will turn until it reaches a position where the weight does not cause moment about the pivot.
(stage of equilibrium)



Stability (definition)

Stability refers to the ability of an object to return to its original position when it is displaced.

Types of Stability

1. Stable Equilibrium

- An equilibrium whereby object will return to its original state when displaced slightly.

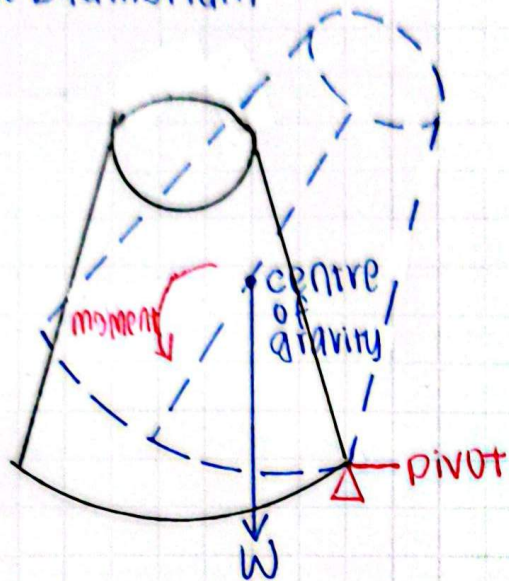
2. Unstable Equilibrium

- An equilibrium whereby object will move further away from its original position when displaced slightly.
(i.e. topple over)

3. Neutral Equilibrium

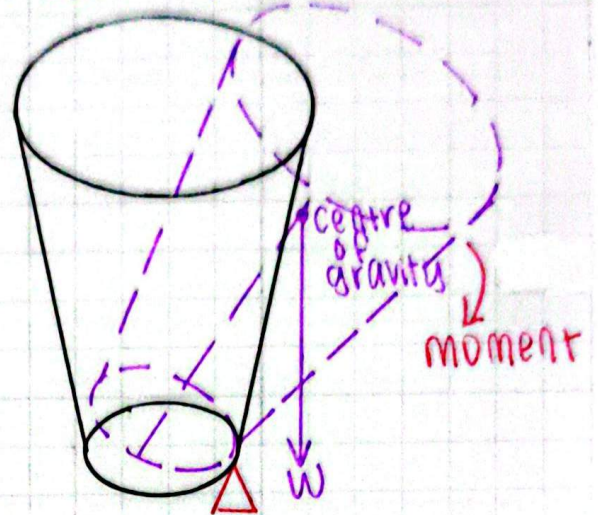
- An equilibrium whereby object will remain at wherever it is displaced.

Stable Equilibrium



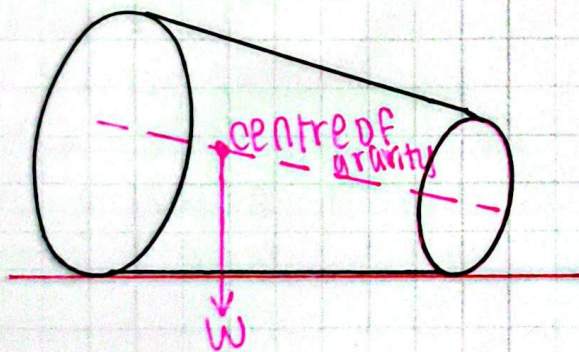
The restoring moment causes the body to return to its original position.

Unstable Equilibrium



The overturning moment causes the body to further displace from its original position.

Neutral Equilibrium



There is no turning effect due to the weight on the body. The line of action of weight always passes through the point of contact.

Making a body more stable

1. Lower centre of gravity (Heavy base)
2. Increase area of base

These ensure that the weight of the object through the centre of gravity falls within its base so it can have stable equilibrium.